

Data-Quality & EDA Insight Memo

Project FORESIGHT · Demand & Inventory Intelligence · NorthBay Living

Client	NorthBay Living	Prepared by	Data Scientist, Zidio engagement
Data window	2024-01-01 to 2025-12-31	SKUs covered	200
Records analyzed	136,175	Deliverable	D2 Memo

This memo summarizes the condition of the data NorthBay Living provided, the fixes applied before any modelling took place, and the demand patterns that fell out of the exploratory analysis. It is written for the Operations, Merchandising, and Finance teams — no code or technical detail is included here; the pipeline itself is documented separately in the project repository.

1. Data-Quality Findings

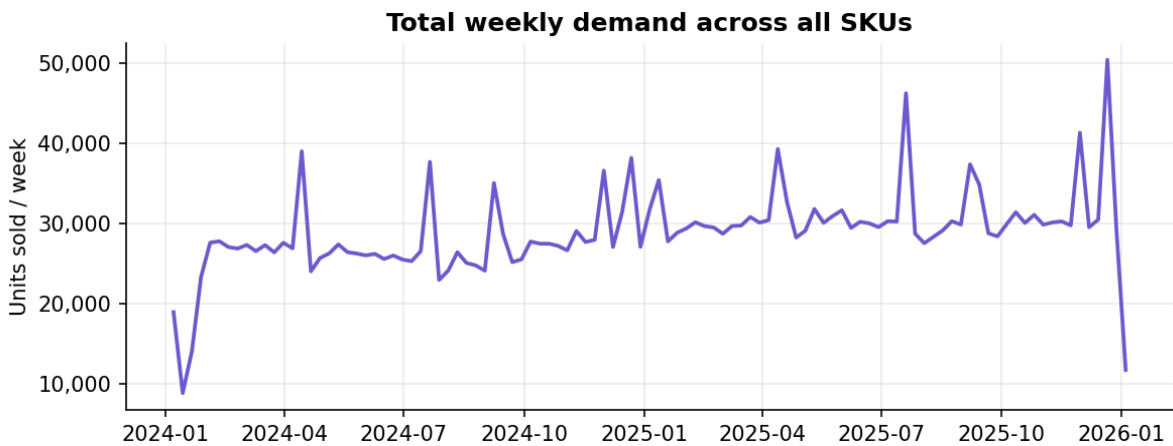
The four raw extracts (sales, product master, calendar, inventory) arrived with the kind of issues typical of spreadsheet-driven operations. Every issue below was found programmatically and corrected before analysis — nothing was hidden or silently dropped.

Data source	Issue found	How it was handled
Product master	Duplicate product entries	Removed exact duplicate rows, kept first occurrence
Product master	Inconsistent category names (e.g. "decor", "DECOR", "Home Decor", "Décor")	Standardized to one consistent label per category
Product master	Missing product cost for a few items	Filled using the typical (median) cost for that item's category
Product master	One product price entered ~100x too high	Corrected to a realistic value based on category pricing
Sales history	Duplicate sales records	Removed duplicates for the same product/day
Sales history	Dates recorded in three different formats	Standardized all dates to one format
Sales history	A small number of negative unit sales (data-entry errors)	Corrected to zero
Sales history	A handful of extreme, implausible sales spikes	Capped at a realistic ceiling per product
Sales history	Missing selling price on some records	Filled from the product's listed price
Inventory records	Duplicate stock snapshots	Removed duplicates for the same product/date
Inventory records	Missing "stock on order" values	Treated as zero (nothing currently on order)

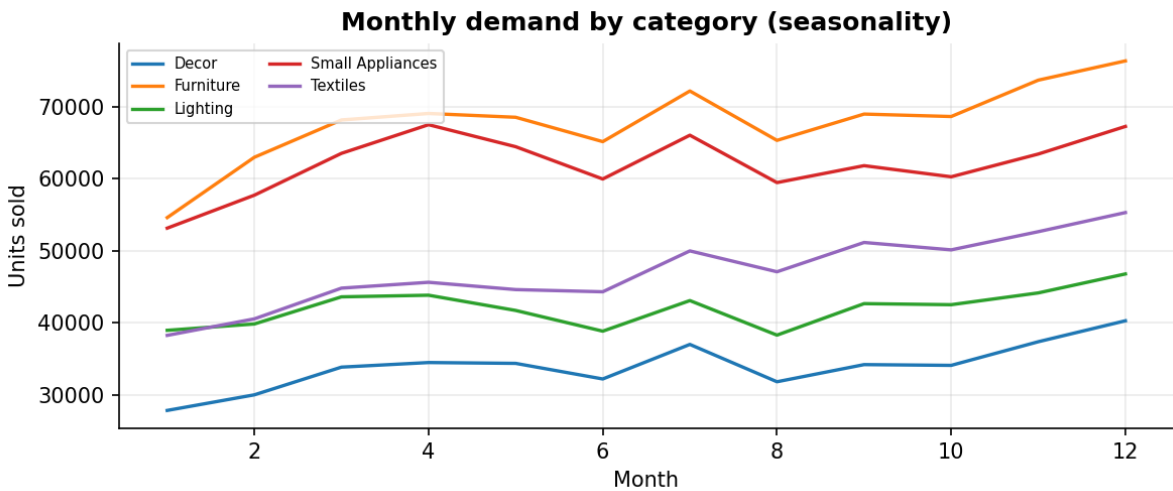
Bottom line: none of these issues were severe enough to lose meaningful data — after cleaning, every one of the 200 products has a usable, consistent sales and inventory history.

2. Demand Patterns

Overall demand has been trending upward over the two-year window, with a clear repeating pattern of short spikes around promotional periods and holidays. The busiest week on record was **2025-12-21** (50,444 units sold across all products), while the quietest was **2024-01-14** (8,831 units) — early in the data window before the business had built up its customer base.

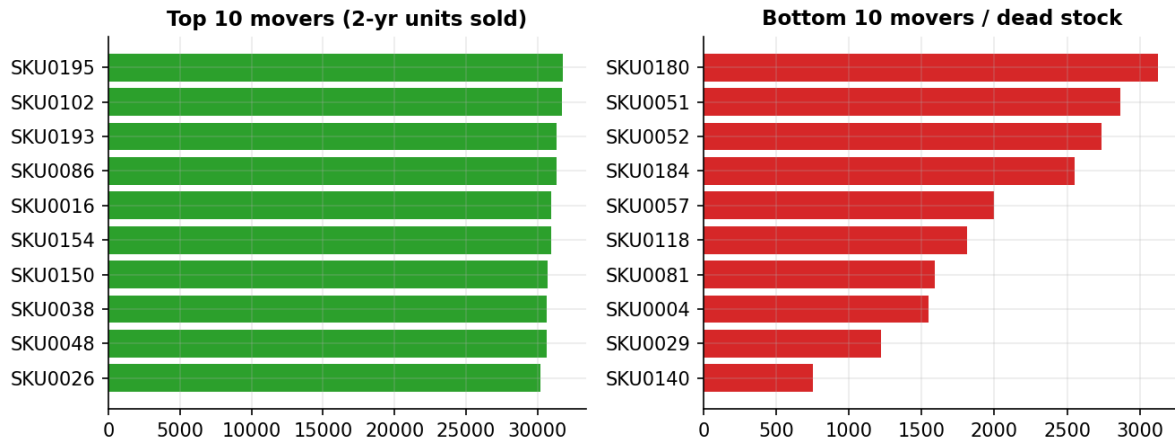


Seasonality differs meaningfully by category. **Furniture** shows the widest swings across the year, meaning stocking plans for that category need to flex with the season rather than use a flat monthly target.



3. Top & Bottom Movers

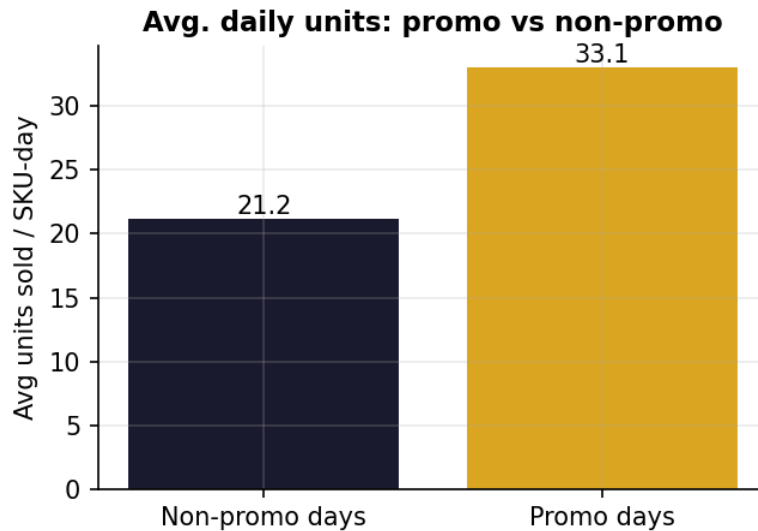
A small group of products consistently outsells the rest, while a similarly small group moves slowly enough to be considered candidates for review (clearance, bundling, or discontinuation). Full SKU-level lists are available in the planning dashboard filters.



No product had effectively zero recent sales — the 'bottom movers' here are still selling, just far more slowly than the top group, which is a useful and honest finding in its own right: it means clearance action should be prioritized by rate of sale, not written off as dead stock.

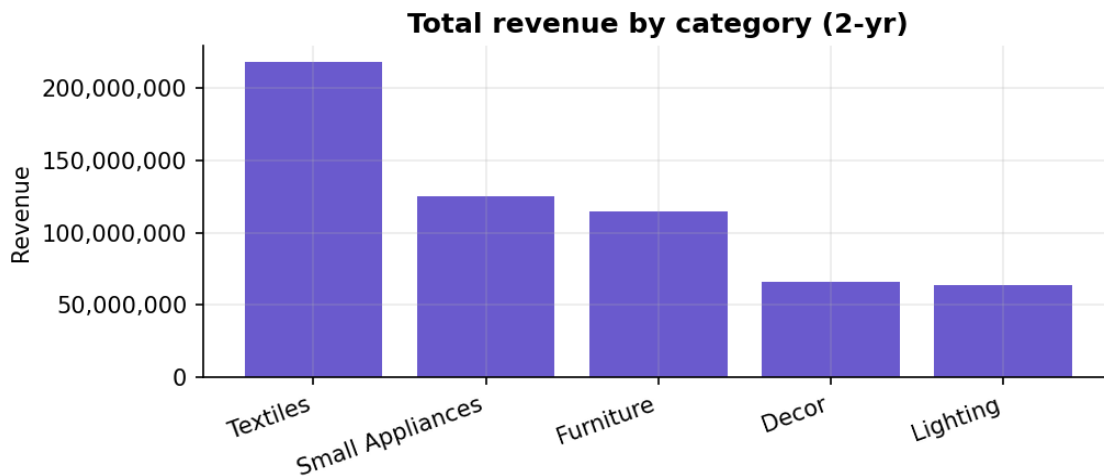
4. Promotion Effect

Promotions clearly move the needle: average daily sales per product are **56% higher** on promo days than on regular days. This lift is already built into the demand forecast so promotional weeks are not systematically under-forecast.



5. Revenue by Category

Textiles is the leading revenue category over the two-year window. Total recorded revenue across all products was approximately INR 587,607,184.



6. Key Business Insights

- 1. Promotions work but are being run without a demand plan behind them.** Promo days sell 56% more per product on average — that lift is now captured in the forecast, so future promo weeks can be stocked for in advance instead of reacted to.
- 2. Seasonality is category-specific, not business-wide.** Furniture swings the most across the year; a single flat reorder rule across all categories will systematically over-stock the flat categories and under-stock the seasonal ones.
- 3. Sales concentration is real but not extreme.** A top group of products consistently outsells the rest, while the slowest movers are still selling — meaning the fix is prioritized promotion/clearance, not blanket discontinuation.
- 4. Data quality was a real (but fixable) risk to trust in the numbers.** Inconsistent category labels alone would have split reporting for the same category into up to five different rows if left uncleaned — this is now standardized so every dashboard figure reconciles correctly.